AP PRECALCULUS

Unit 1 • Lesson 1.6 practice

Lesson 1.6 • Polynomial Functions and End Behavior

4 multiple-choice questions and 1 free-response question

Question 1

$\lim_{x\to\infty}\left(-4x^3+x^2\right)$ is

(A) ∞

(B) $-\infty$

(C) -4

(D) 0

Question 2

Let $p(x)=(2-x)^3$. What is $\lim_{x\to-\infty} p(x)$?

(A) $-\infty$

(B) $+\infty$

(C) 8

(D) 0

Question 3

Which polynomial increases without bound as $x\to\infty$ and as $x\to-\infty$?

(A) x^3+1

(B) $-x^4$

(C) $5x-x^2$

(D) $2x^6-x$

Question 4

The graph of a polynomial falls to the left and rises to the right. Its degree and leading coefficient are

(A) even, positive

(B) even, negative

(C) odd, positive

(D) odd, negative

Question 5 (Free response)

Let $f(x)=-2x^4+3x^3+x-7$ and $g(x)=x^2(3-x)(x+1)^2$.

(a) Write limit statements for the end behavior of f.

(b) Find the degree and leading term of g, and write limit statements for its end behavior.

(c) Could the graph of g have 5 turning points? Explain.